RFBmodel1D

Supplementary Material: Model equations

Properties

$$D_j^{eff} = \varepsilon^{3/2} D_j \quad (E.1)$$

$$u_j = \frac{D_j}{RT} \quad (E.2)$$

$$\sigma_{el}^{eff} = (1-\varepsilon)^{3/2} \sigma_{el} \quad (E.3)$$

Cathode/Membrane Mass and charge balances

Equation	Domain	Note
$V \cdot \frac{\partial c}{\partial t} + A \cdot N_j = 0$	Tank	*1*
$\frac{\varepsilon \partial c_j}{\partial t} + \nabla N_j = a\zeta_i$	Cathode	*2*
$\frac{\varepsilon \partial c_j}{\partial t} + \nabla N_j = 0$	Membrane	*3*
$\nabla i_{el} = j_n$	Cathode	
$\nabla i_{el} = -j_n$	Cathode	

Equations

Equation	Domain	Note
$N_j = -D_j^{eff} \nabla c_j - z_j u_j c_j F \nabla \Phi_{io} + v_{\frac{a}{c}} c_j$	Cathode, Membrane	-
$\sum z_j c_j = 0$	Cathode	*4*
$i_{el} = -\sigma_{el} \cdot \nabla \Phi_{el}$	Cathode	
$i_{el} = -\kappa_{io} \cdot \nabla \Phi_{el}$	Cathode	
$j_{nc} = aFk_c(c_{V4})^{\alpha_+}(c_{V5})^{1-\alpha_+}\left(\left(\frac{c_s^{V4}}{c^{V4}}\right)\exp\left((1-\alpha_+)f\eta_c\right) - \left(\frac{c_s^{V5}}{c^{V5}}\right)\exp\left((-\alpha_+)f\eta_c\right)\right)$	Cathode	
$c_s^{V4} = (\Psi_4 \cdot c_{V5} + (1+\Psi_4)\cdot c_{V4})/(1+\Psi_3+\Psi_4)$ $c_s^{V5} = (\Psi_3 \cdot c_{V4} + (1+\Psi_3)\cdot c_{V5})/(1+\Psi_3+\Psi_4)$ $\Psi_3 = \frac{k}{B_0}\frac{c_{V5}^{\alpha}}{c_{V4}^{1-\alpha}} \cdot \exp\left((1-\alpha)f\eta\right)$ $\Psi_4 = \frac{k}{B_0}\frac{c_{V4}^{1-\alpha}}{c_{V5}^{\alpha}} \cdot \exp\left(-\alpha f\eta\right)$	Cathode	

Boundary Conditions

The finite volume discretisation treats boundary conditions as flux contributions at the domain boundaries. Specifically, the fluxes through the leftmost and rightmost faces of each domain are prescribed to enforce the corresponding Dirichlet or Neumann conditions at the boundary nodes.

Equation	Domain	*Note*
$N_{BC} = -D_{j,m}\frac{(c_{cath} - c_{BC})}{\frac{\Delta x}{2}} - z_j u_j \frac{(c_{cath} + c_{BC})}{2} F \cdot \nabla \Phi_{io}$	CC/Cathode Interface	*1*
$N_{j \neq H+} = 0$ $N_{H+} = -D_{H+}\frac{(c_{BC} - c_{cath})}{\frac{\Delta x}{2}} - zu\frac{(c_{BC} + c_{cath})}{2} F \cdot \frac{\Phi_{\mathrm{io}}^{BC} - \Phi_{\mathrm{io}}^{\mathrm{cath}}}{\frac{\Delta x}{2}}$	Cathode/Membrane Interface. Cathode side.	*2*
$i_{io,cath} = -\kappa_{io,cath} \cdot \frac{\boldsymbol{\Phi}_{\mathbf{io}}^{\boldsymbol{BC,cath}} - \Phi_{\mathrm{io}}^{\mathrm{cath}}}{\frac{\Delta x}{2}}$ $i_{io,mem} = -\kappa_{io,mem} \cdot \boldsymbol{\nabla \Phi_{io}}$ $i_{io,cath} = F \cdot (-D_{H+}\frac{(\boldsymbol{c_{BC}} - c_{cath})}{\frac{\Delta x}{2}} - zu\frac{(\boldsymbol{c_{BC}} + c_{cath})}{2} F \cdot \frac{\Phi_{\mathrm{io}}^{BC} - \Phi_{\mathrm{io}}^{\mathrm{cath}}}{\frac{\Delta x}{2}})$	Cathode/Membrane Interface.	*3*

Source Terms

Term	Positive electrode
$\zeta_{VO^{2+}}$	$-\frac{j_n}{aF}$
$\zeta_{VO_2^+}$	$\frac{j_n}{aF}$
ζ_{H^+}	$2\frac{j_n}{aF} - \frac{s_d}{a}$
$\zeta_{HSO_4^-}$	$\frac{s_d}{a}$

0D Mass balances

Equation	**Domain**	**Note**
$\frac{dc_{vw,ACL}}{dt} \cdot l_{ACL} = J_{iw,vw} - J_{vw} + R_{w,ACL}$	ACL, AGDL	*5*

$\frac{dc_{vw,AGDL}}{dt} \cdot l_{AGDL} = J_{vw} - J_{vw,ch} + R_{w,AGDL}$		
$\frac{dc_{H2,ACL}}{dt} \cdot l_{ACL} = -J_{H2} + S_{BV}$ $\frac{dc_{H2,AGDL}}{dt} \cdot l_{AGDL} = J_{H2} - J_{H2,ch}$	ACL, AGDL	
$\frac{ds_{ACL}}{dt} \cdot l_{ACL} \cdot \varepsilon \cdot \frac{\rho_{lw}}{MW} = J_{iw,lw} - J_{lw} - R_{w,ACL}$ $\frac{ds_{AGDL}}{dt} \cdot l_{AGDL} \cdot \varepsilon \cdot \frac{\rho_{lw}}{MW} = J_{lw} - J_{lw,ch} - R_{w,AGDL}$	ACL, AGDL	
$R_w = -k_{cond} \cdot l \cdot \varepsilon(1-s) \cdot \frac{c_{vw}}{c_{vw}+c_{H2}} \cdot (c_{vw} - c_{sat})$ for $c_{vw} > c_{sat}$ $R_w = k_{evap} \cdot l \cdot \varepsilon \left(\frac{s}{s-s_{im}}\right) \cdot \frac{\rho_{lw}}{MW} \cdot \frac{p_{vw}-p_{sat}}{101325}$ for $c_{vw} < c_{sat}$	ACL, AGDL	
$i_{cath} = \frac{j_0 n F \eta_{an}}{RT}$	ACL	

Potential equations

$\eta_c = \Phi_s - \Phi_e - E_c$	Cathode
$\eta_a = \Phi_s - \Phi_e - E_a$	*Anode*
$E_c = E_{0,c} + \frac{RT}{F} \ln\left(\frac{c_{VO_2^+} c_{H^+}^2}{c_{VO^{2+}}}\right)$	Cathode
$E_a = E_{0,a} + \frac{RT}{F} \ln\left(\frac{c_{H+}^2}{p_{H2}}\right)$	*Anode*

State variables

c_j : Concentration of species j / mol·m^{-3}

φ_{io} : $Ionic\ potential\ /\ V$

φ_{el} : $Electronic\ potential\ /\ V$

Potential and Transport

u_j : Ionic mobility / m^2·V^{-1}·s^{-1}

z_j : Charge number of species j / -

N_j : Molar flux of species j / mol·m^{-2}·s^{-1}

D_j : Molecular diffusion coefficient / m^2·s^{-1}

D_j^{eff} : Effective diffusion coefficient / m^2·s^{-1}

i_{io} : $Ionic\ current\ density\ /\ A \cdot m^{-2}$

i_{el} : $Electronic\ current\ density\ /\ A \cdot m^{-2}$

σ_s : $Solid-phase\ conductivity\ /\ S \cdot m^{-1}$

η : $Overpotential, anode\ or\ cathode\ /\ V$

Other

ε : $Porosity\ /\ -$

a : $Specific\ surface\ area\ /\ m^2 \cdot m^{-3}$

v : $Superficial\ velocity\ /\ m \cdot s^{-1}$

j_n : $Volumetric\ reaction\ current\ density\ /\ A \cdot m^{-3}$

k : $Reaction\ rate\ constant\ /\ m \cdot s^{-1}$

α : $Charge\ transfer\ coefficient\ /\ -$

E_{eq} : $Equilibrium\ potential\ /\ V$

Notes for Mass Balances

1. Tank Mass balance.

$$V \cdot \frac{\partial c}{\partial t} = N_{in} - N_{out} + R$$

The tank and channel concentrations are assumed equal. The interfacial mass flux is computed from the gradient between the tank and the first cathode node, after which the cathode boundary condition is updated accordingly.

$$N_j = -D\frac{c_{cathode} - c_{tank}}{\frac{\Delta x}{2}} - z_j F \mu_j \cdot \frac{c_{cathode} + c_{tank}}{2} \cdot \frac{d\Phi}{dx} + v \cdot \frac{c_{cathode} + c_{tank}}{2}\ [\frac{mol}{m^2 s}]$$

This flux goes from tank to cathode, so it can be replaced with N_out. There is no reaction.

2. Cathode species mass balance.
The balance for vanadium species is:

$$\frac{\varepsilon \partial c_j}{\partial t} + \nabla N_j = \pm \frac{j_n}{F}$$

Where j_n is the volumetric term for Butler-Volmer reaction. For discharge its value is negative. So, it will have a positive sign for V4 (produced) and a negative one (consumed) for V5.

The balance for H+ and HSO4- depends on empiric equation of acid dissociation, for example, for proton:

$$\frac{\varepsilon \partial c_j}{\partial t} + \nabla N_j = 2\frac{j_n}{F} - s_d$$

Where:

$$s_d = k_d \cdot \left(\frac{c_{H+} - c_{HSO4-}}{c_{H+} + c_{HSO4-}} - \mathrm{B} \right)$$

3. Membrane mass balance.
   There is only proton crossover, so membrane conductivity should be constant, and membrane ionic potential should be linear.
4. In cathode, SO4+ is calculated using electroneutrality condition. In membrane, only proton is present, so their concentration is constant.

$$c_{H+} = c_f$$

Notes for Boundary Conditions

1. Boundary Condition of ionic potential in CC/Cathode interface is always Neumann with gradient equal to 0. Boundary conditions of species concentrations are always Dirichlet and equal to tank concentration, so this face flux is calculated as.

$$N_{BC} = -D_{j,m} \frac{(c_{cath} - c_{BC})}{\frac{\Delta x}{2}}$$

2. At the cathode/membrane interface, Dirichlet boundary conditions are enforced for both the ionic potential and the proton concentration. All other species are assigned to a Neumann condition, corresponding to zero flux across the interface.

3. At the cathode/membrane interface, continuity of ionic current is enforced, where the ionic current equals the total Butler-Volmer reaction current integrated across all cathode nodes. On the cathode side, this current is used to calculate the ionic potential to the interface face, which is then imposed as a Dirichlet boundary condition. On the membrane side, the same current prescribes the potential gradient as a Neumann condition.

   Given that the membrane is only permeable to protons, the proton flux is determined by the ionic current via Faraday's law, from which the interfacial proton concentration at the cathode side is imposed as a Dirichlet condition.
4. The channel is modelled as a stationary boundary with fixed conditions: zero reduced liquid saturation, fully saturated vapor (RH = 100%), and an H2 concentration of 42 mol/m3. Given that both the ACL and AGDL are discretized as single 0D nodes, mass fluxes are evaluated as finite differences between the mean concentration of each domain, using half the sum of their respective thicknesses as the effective transport length. For instance, the vapor water flux between the ACL and AGDL is

$$J_{vw} = -D_{vw} \frac{\left(c_{vw,AGDL} - c_{vw,ACL}\right)}{l_{\frac{ACL}{AGDL}}}$$